

Research Focus

Genomic Data & Information Theory: Applying AI/ML algorithms to extract insights from protein, DNA, and RNA sequence data; developing computational frameworks at the intersection of information theory and bioinformatics.

Protein Design (Govt. of India Project): Contributing to large-scale protein design for **NS5 and Dengue vaccine** development via computational modeling, deep learning, and molecular dynamics.

Core AI/ML & Agentic Systems: Developing foundational algorithms for **agentic AI**, **AI safety/hardness**, and **retrieval-augmented generation** systems.

Publications

AgriIR: Scalable Framework for Domain-Specific Knowledge Retrieval
ECIR 2026
S. Banerji Seal, A. Poddar, A. Mishra, D. Roy [DOI](#)

Hierarchical Opinion Classification using LLMs
FIRE 2025
S. Banerji Seal, A. Mishra, U. Ghosh [Link](#)

Query Reformulation & Layered Retrieval for ToT
TREC 2024
S. Banerji Seal, S. Adhikary, S. Sar, D. Roy [Link](#)

Computational Modeling of [VO(SALIEP)(DTP)]
To be submitted
S. Banerji Seal, Dr. S. Roy. DFT methods for MO energy calculations.

Startups

Co-Founder & Lead Developer, iFiNN
2025–Present
MeitY GENESIS-funded AI-fintech startup. Smart trading alerts for stocks, crypto, mutual funds. [Website](#)

Co-Founder & CTO, UnderWater AI
2025–Present
MeitY GENESIS-funded DeepTech startup. Deep learning for underwater image enhancement. [Website](#)

Leadership & Community

- **President**, Slashdot – Programming & Design Club, IISER-K (2025–present)
- **Track Co-organizer, FIRE 2026** – **SYCO PHANCY**: Explainable AI in Legal Reasoning (IISER-K, UBO France, UvA Netherlands)
- **Organizer**, Qiskit Fallfest 2025 (IBM Quantum, 1 of 150 globally)
- **Office Bearer**, Valence Chemistry Club, IISER-K
- COVID-19 relief coordinator; Ek-Pehal mentoring initiative

Hackathons & Awards

UIDAI Data Hackathon 2026 – 1st Prize (Rs. 2L)
Jan 2026
National hackathon by UIDAI/NIC/MeitY. Data-driven Aadhaar enrolment analysis. [Link](#)

Capital One Launchpad – Top 14 / 5,073 Teams
2025
IndicAgri: multi-modal RAG for Indian agriculture, 20+ languages (Team Fibonacci). [GitHub](#)

StatusCode1 – 1st Rank (GIAN Track)
2024
AI search engine for GIAN’s Abandoned US Patents. Nomic Embeddings + similarity search. [GitHub](#)

StatusCode0 – 1st Rank (MATLAB Track)
2023
Domestic waste data analysis tool, IIIT-Kalyani.

Smart India Hackathon – Finalist
2024
Problem IDs: SIH1701 (Context-Aware AI for Courts), SIH1734 (Deep Learning for Air Quality).

National Competitions

- **ChemEnigma** (1st) – 72-hr chemistry contest, IISc Bangalore (2025)
- **All Bengal Chemistry Quiz** (2nd RU), Presidency University (2025)
- **Mimansa** (Zonal Topper) – Math, IISER Pune (2024)
- **NAEST** (Zonal RU) – Experimental Skills, IIT Kanpur (2023)
- **JEE Advanced** (Top 0.1%) | **IAT** (Top 0.06%) | **WBJEE** (Top 0.05%)

Technical Skills

- **AI/ML:** PyTorch, TensorFlow, HuggingFace, LangChain, Gemma, Qwen, DeepSeek, LLaMA
- **IR & RAG:** BM25, RRF, FAISS, ChromaDB, Dense+Sparse Retrieval, Vector DBs
- **Bioinformatics:** LAMMPS, Gaussian, PyMol, ChimeraX, DFT, AutoDock Vina
- **Agentic AI:** Autonomous agents, sycophancy detection, legal AI reasoning
- **Languages:** Python, C/C++, Java, Rust, Fortran
- **Web/Dev:** Django, Flask, React, Three.js, Docker, Git, Linux

Open-Source Libraries

Fernholz SPT Library
Ongoing
Stochastic Portfolio Theory: diversity-weighted portfolios, relative arbitrage. [GitHub](#)

LAMMPS Data Web Viewer
2025
3D molecular dynamics visualization (React, Three.js, Flask). [GitHub](#) · [Live](#)

Education

- **IISER Kolkata** – BS-MS (Chemistry Major, CS Minor), CGPA: 8.2 (2022–2027)
- **Calcutta University** – B.Sc Honours Physics, CGPA: 8.308 (2021–22)
- **Reliance Foundation Scholar** (2023) | **Best Young Scientist Speaker** (2019)